Use Cases

for

HDB Resale Flat Search and Analysis System

Version 1.2

Prepared by Team ABCDE

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Revision History

Name	Date	Reason For Changes	Version
Roen, Aleesha, Manwen, Jing Yi, Junyao	6/9/2026	Initial Edit of the Document	1.0
Roen, Aleesha, Manwen, Jing Yi, Junyao	21/9/2026	Adding Extra Use Cases	1.1
Roen, Aleesha, Manwen, Jing Yi, Junyao	1/10/2026	Lab 3: completed use case model. Fixed include and extend directions, aligned descriptions with SRS v1.2, added UC-13 to UC-15 and the use case diagram. Update on 9/11/2026: UC-02 now names OneMap as the only travel-time source, and UC-04 says recent sale details instead of listing details	1.2

Guidance for Use Case Template

Document each use case using the template shown in the Appendix. This section provides a description of each section in the use case template.

1. Use Case Identification

1.1. Use Case ID

Give each use case a unique numeric identifier, in hierarchical form: X.Y. Related use cases can be grouped in the hierarchy. Functional requirements can be traced back to a labelled use case.

1.2. Use Case Name

State a concise, results-oriented name for the use case. These reflect the tasks the user needs to be able to accomplish using the system. Include an action verb and a noun. Some examples:

- View part number information.
- Manually mark hypertext source and establish link to target.
- Place an order for a CD with the updated software version.

1.3. Use Case History

1.3.1 Created By

Supply the name of the person who initially documented this use case.

1.3.2 Date Created

Enter the date on which the use case was initially documented.

1.3.3 Last Updated By

Supply the name of the person who performed the most recent update to the use case description.

1.3.4 Date Last Updated

Enter the date on which the use case was most recently updated.

2. Use Case Definition

2.1. Actor

An actor is a person or other entity external to the software system being specified who interacts with the system and performs use cases to accomplish tasks. Different actors often correspond to different user classes, or roles, identified from the customer community that will use the product. Name the actor(s) that will be performing this use case.

2.2. Description

Provide a brief description of the reason for and outcome of this use case, or a high-level description of the sequence of actions and the outcome of executing the use case.

2.3. Preconditions

List any activities that must take place, or any conditions that must be true, before the use case can be started. Number each precondition. Examples:

1. User's identity has been authenticated.
2. User's computer has sufficient free memory available to launch task.

2.4. Postconditions

Describe the state of the system at the conclusion of the use case execution. Number each postcondition. Examples:

1. Document contains only valid SGML tags.
2. Price of item in database has been updated with new value.

2.5. Priority

Indicate the relative priority of implementing the functionality required to allow this use case to be executed. The priority scheme used must be the same as that used in the software requirements specification.

2.6. Frequency of Use

Estimate the number of times this use case will be performed by the actors per some appropriate unit of time.

2.7. Flow of Events

Provide a detailed description of the user actions and system responses that will take place during execution of the use case under normal, expected conditions. This dialog sequence will ultimately lead to accomplishing the goal stated in the use case name and description. This description may be written as an answer to the hypothetical question, "How do I <accomplish the task stated in the use case name>?" This is best done as a numbered list of actions performed by the actor, alternating with responses provided by the system.

2.8. Alternative Flows

Document other, legitimate usage scenarios that can take place within this use case separately in this section. State the alternative course, and describe any differences in the sequence of steps that take place. Number each alternative course using the Use Case ID as a prefix, followed by “AC” to indicate “Alternative Course”. Example: X.Y.AC.1.

2.9. Exceptions

Describe any anticipated error conditions that could occur during execution of the use case, and define how the system is to respond to those conditions. Also, describe how the system is to respond if the use case execution fails for some unanticipated reason. Number each exception using the Use Case ID as a prefix, followed by “EX” to indicate “Exception”. Example: X.Y.EX.1.

2.10. Includes

List any other use cases that are included (“called”) by this use case. Common functionality that appears in multiple use cases can be split out into a separate use case that is included by the ones that need that common functionality.

2.11. Special Requirements

Identify any additional requirements, such as nonfunctional requirements, for the use case that may need to be addressed during design or implementation. These may include performance requirements or other quality attributes.

2.12. Assumptions

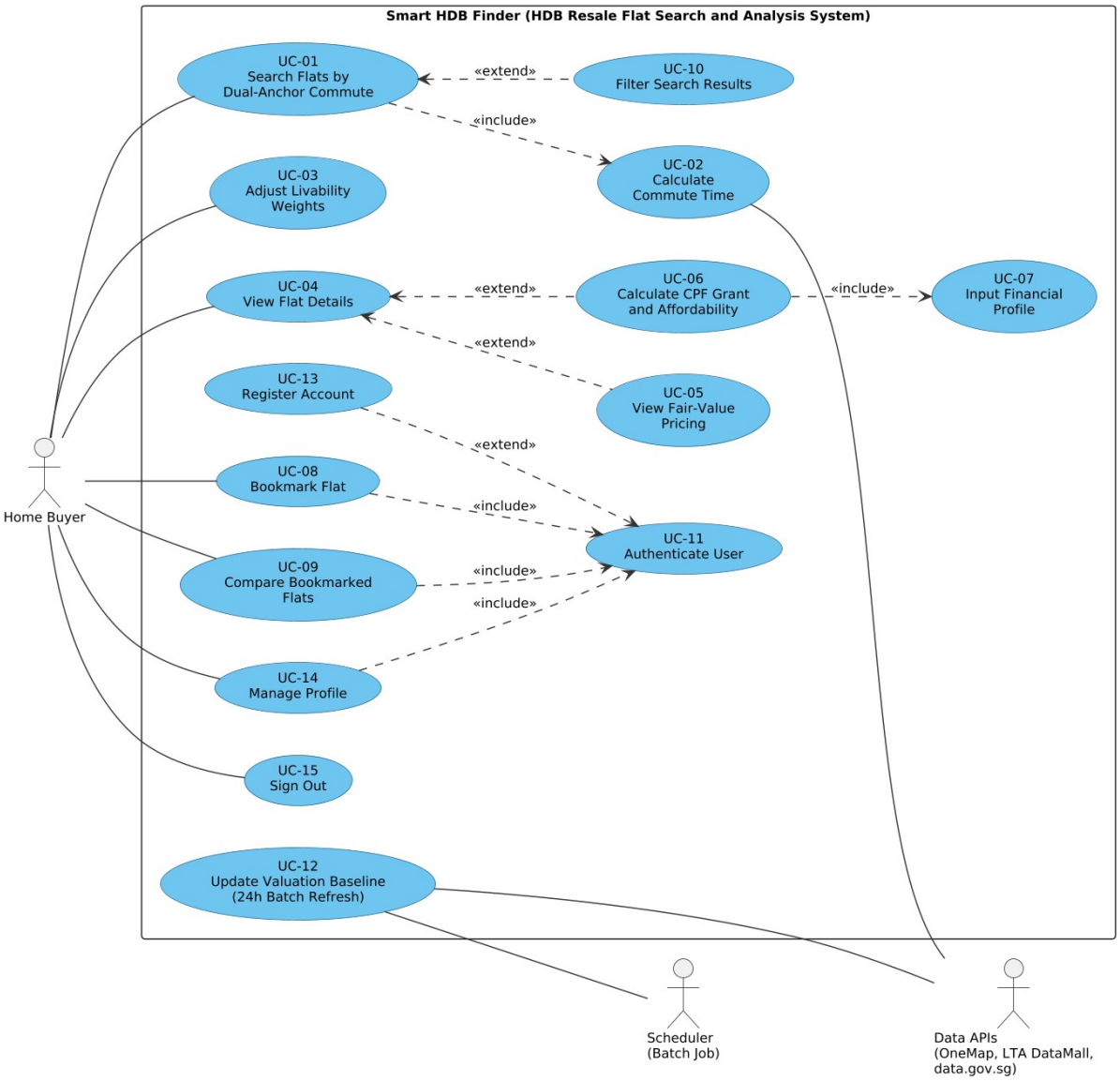
List any assumptions that were made in the analysis that led to accepting this use case into the product description and writing the use case description.

2.13. Notes and Issues

List any additional comments about this use case or any remaining open issues or TBDs (To Be Determineds) that must be resolved. Identify who will resolve each issue, the due date, and what the resolution ultimately is.

Use Case Diagram

The use case diagram below shows the complete use case model for Lab 3. It has three actors and fifteen use cases. The Home Buyer is the main actor. The Data APIs (OneMap, LTA DataMall and data.gov.sg) and the Scheduler are secondary actors that support commute calculation and the daily valuation refresh.



Use Case Diagram (version 1.2)

Changes from Version 1.1

The following changes were made so that the use case model is complete and consistent with the SRS:

1. The extend relationships for View Fair-Value Pricing (UC-05) and Calculate CPF Grant and Affordability (UC-06) now point to View Flat Details (UC-04). In UML, the arrow goes from the optional extension to the base use case.

2. Filter Search Results (UC-10) is now an extension of UC-01 instead of an include, since filtering is optional.
3. Compare Bookmarked Flats (UC-09) is now a standalone use case that includes Authenticate User. It no longer extends Bookmark Flat.
4. Three new use cases were added: Register Account (UC-13), Manage Profile (UC-14) and Sign Out (UC-15). These cover the Login, Sign-Up and Profile screens in the dialog map.
5. Descriptions were aligned with the SRS. The livability factors are remaining lease, crime rate, residential density and air quality. The financial profile fields are household income, marital status and parental address.
6. UC-05 now reads from the stored valuation baseline maintained by UC-12, so it no longer calls data.gov.sg directly.
7. Extension use cases are not linked to the Home Buyer directly, because they are triggered from inside their base use case.

Use Case Summary

ID	Use Case Name	Actor(s)	Priority	Related SRS Requirements
UC-01	Search Flats by Dual-Anchor Commute	Home Buyer	High	FR-1 to FR-4, FR-17 to FR-20, FR-22, FR-23, FR-31
UC-02	Calculate Commute Time	Data APIs	High	FR-3, NFR-3
UC-03	Adjust Livability Weights	Home Buyer	High	FR-5 to FR-8, FR-21
UC-04	View Flat Details	Home Buyer	High	FR-23, NFR-3
UC-05	View Fair-Value Pricing	Home Buyer	High	FR-9, FR-32
UC-06	Calculate CPF Grant and Affordability	Home Buyer	High	FR-11 to FR-13
UC-07	Input Financial Profile	Home Buyer	High	FR-10, NFR-6
UC-08	Bookmark Flat	Home Buyer	Medium	FR-14, FR-15, FR-21
UC-09	Compare Bookmarked Flats	Home Buyer	Medium	FR-16
UC-10	Filter Search Results	Home Buyer	Medium	FR-18, FR-20, FR-22
UC-11	Authenticate User	Home Buyer	High	FR-24 to FR-26
UC-12	Update Valuation Baseline (24h Batch Refresh)	Scheduler, Data APIs	Medium	FR-28 to FR-30, NFR-5
UC-13	Register Account	Home Buyer	High	FR-27
UC-14	Manage Profile	Home Buyer	Medium	FR-33, FR-34
UC-15	Sign Out	Home Buyer	Medium	FR-35

Use Case ID:	UC-01		
Use Case Name:	Search Flats by Dual-Anchor Commute		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	6 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Allows the Home Buyer to search for HDB resale flats using one or two workplace anchors (for example, a workplace and a family home). The system ranks matching flats by an overall match score that combines commute convenience to the anchors with each flat's Livability Index.
Preconditions:	Home Buyer has launched the application. Signing in is not required to search.
Postconditions:	A ranked list and a map view of matching flats are displayed to the Home Buyer. The search result set is kept by the system so it can be re-weighted (UC-03) or filtered (UC-10) without querying the Data APIs again.
Priority:	High
Frequency of Use:	Frequent. Used at the start of most sessions.
Flow of Events:	1. Home Buyer enters Workplace 1 and, optionally, Workplace 2. 2. Home Buyer selects preferred transport modes, budget range, flat types and towns. 3. Home Buyer clicks "Search Flats". 4. System checks that Workplace 1 is provided and that each address can be resolved to map coordinates. 5. System retrieves candidate flats that match the budget, flat type and town criteria. 6. System invokes Calculate Commute Time (UC-02) for each candidate flat against each anchor. 7. System computes a commute score and a Livability Index (using the current weights) for each flat. 8. System ranks the flats by overall match score. 9. System displays the ranked results as a list and a map, showing each flat's location, flat type, price, Livability Index and valuation label.
Alternative Flows:	UC-01.AC.1: If the Home Buyer supplies only Workplace 1, the system computes a single-anchor commute score. UC-01.AC.2: If the Home Buyer is signed in, the system pre-loads their last-used weight profile before the search (FR-21).
Exceptions:	UC-01.EX.1: If Workplace 1 is missing, the system prompts the Home Buyer to enter it and does not run the search (FR-19). UC-01.EX.2: If an anchor address cannot be resolved, the system shows an error next to that field and asks the Home Buyer to re-enter it.

	UC-01.EX.3: If an external API fails, the system uses the cached commute time and shows a warning (UC-02). Only if there is neither a live nor a cached commute time for a flat does the system leave that flat out of the ranked results and show a notice. UC-01.EX.4: If no flats match, the system shows a “No matching flats” message and suggests which filters to relax (FR-20).
Includes:	Includes: Calculate Commute Time (UC-02). Extended by: Filter Search Results (UC-10).
Special Requirements:	Results for up to 500 flats must be returned within 3 seconds at the 95th percentile (NFR-1), using cached and pre-computed commute times. If a Data API is slow or down, cached data is used with a warning (NFR-3).
Assumptions:	Anchor addresses can be resolved to valid map coordinates by OneMap. Candidate flats come from recent resale transactions on data.gov.sg, since there is no public API for live listings.
Notes and Issues:	The exact overall match score formula is listed as TBD-2 in the SRS (Appendix C).

Use Case ID:	UC-02		
Use Case Name:	Calculate Commute Time		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	6 September 2026	Date Last Updated:	1 October 2026

Actor:	Data APIs (OneMap)
Description:	Computes the travel time between a flat and an anchor location using external mapping and transport data, with a local cache so repeated requests are fast. This use case is included by Search Flats by Dual-Anchor Commute.
Preconditions:	A valid flat location, anchor location and transport mode are supplied by the calling use case.
Postconditions:	A commute time in minutes is returned to the calling use case and stored in the commute cache.
Priority:	High
Frequency of Use:	Invoked automatically for every candidate flat whenever UC-01 is performed.
Flow of Events:	1. Calling use case supplies a flat location, an anchor location and the selected transport mode. 2. System checks the commute cache for a recent travel time for this flat, anchor and mode. 3. If no recent cached value exists, system requests route and travel-time information from the Data APIs. 4. System stores the returned travel time in the commute cache. 5. System returns the commute time to the calling use case.
Alternative Flows:	UC-02.AC.1: If a recent cached value exists, the system returns it without calling the Data APIs.
Exceptions:	UC-02.EX.1: If the Data APIs fail or take longer than 2.0 seconds, the system falls back to the most recent cached value within 500 ms and marks it as possibly outdated (NFR-3). UC-02.EX.2: If the Data APIs fail and no cached value exists, the system returns an error to the calling use case.
Includes:	None.
Special Requirements:	Depends on the availability and response time of OneMap. All calls use a 2.0 second timeout.
Assumptions:	The Data APIs provide accurate and reasonably current travel-time information.
Notes and Issues:	None at this time.

Use Case ID:	UC-03		
Use Case Name:	Adjust Livability Weights		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	6 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Allows the Home Buyer to change how much each of the four livability factors (remaining lease, crime rate, residential density and air quality) counts towards the Livability Index, and see the results re-ranked.
Preconditions:	Home Buyer is on the Search screen or the Results screen.
Postconditions:	The four weights add up to 100%. If a result list exists, the Livability Index of every flat is recalculated and the list is re-ranked. If the Home Buyer is signed in, the weights are saved as their last-used weight profile.
Priority:	High
Frequency of Use:	Occasional. Used when refining search results.
Flow of Events:	1. Home Buyer moves one of the four livability sliders. 2. System checks that the new value is between 0% and 100%. 3. System adjusts the other three weights in proportion so the total stays at 100% (FR-6). 4. System recalculates the Livability Index for every flat in the current result set. 5. System re-ranks and re-displays the result list without a full page reload. 6. If the Home Buyer is signed in, system saves the weights as the last-used weight profile (FR-21).
Alternative Flows:	UC-03.AC.1: Home Buyer resets the weights to the system defaults. UC-03.AC.2: If no search has been run yet, the system stores the weights and applies them to the next search. Steps 4 and 5 are skipped.
Exceptions:	UC-03.EX.1: If a weight value is negative or above 100%, the system shows a validation error and keeps the previous weights.
Includes:	None.
Special Requirements:	Re-ranking must finish within the NFR-1 target (3 seconds for up to 500 flats) and must not call the Data APIs again.
Assumptions:	Default livability weights have been defined by the system (TBD-3 in the SRS).
Notes and Issues:	None at this time.

Use Case ID:	UC-04		
Use Case Name:	View Flat Details		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	6 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Allows the Home Buyer to view detailed information about a selected flat, including its recent sale details, commute breakdown, Livability Index breakdown, valuation label and nearby schools.
Preconditions:	Home Buyer has a list of search results or bookmarks displayed.
Postconditions:	Detailed information for the selected flat is displayed to the Home Buyer.
Priority:	High
Frequency of Use:	Frequent. Used for most flats of interest.
Flow of Events:	1. Home Buyer selects a flat card or a map pin. 2. System retrieves the flat's address, flat type, floor area, storey range, remaining lease and price. 3. System displays the commute breakdown for each anchor, the Livability Index with its four factor scores, the valuation label and nearby schools (for information only). 4. System shows options to view fair-value pricing, open the CPF calculator, bookmark the flat or compare it.
Alternative Flows:	UC-04.AC.1: Home Buyer opens the flat from the Bookmarks screen instead of the Results screen. If there is no search in the current session, the commute breakdown shows as not available.
Exceptions:	UC-04.EX.1: If flat details cannot be retrieved because an external data source is unavailable, the system falls back to the most recent cached data and tells the Home Buyer the data may be outdated (NFR-3).
Includes:	None. Extended by: View Fair-Value Pricing (UC-05) at the "Fair-value pricing" extension point, and Calculate CPF Grant and Affordability (UC-06) at the "CPF calculator" extension point.
Special Requirements:	None identified.
Assumptions:	Flat details are available from the system's flat database.
Notes and Issues:	Version 1.1 wrote that this use case extends View Fair-Value Pricing. The direction was reversed. View Fair-Value Pricing is the optional extension of this use case.

Use Case ID:	UC-05		
Use Case Name:	View Fair-Value Pricing		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	6 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Displays a fair-value price estimate for a selected flat, based on the rolling 6-month estate average from historical resale transactions. This use case extends View Flat Details.
Preconditions:	Home Buyer is viewing the details of a selected flat. A valuation baseline exists for the flat's town and flat type (maintained by UC-12).
Postconditions:	The valuation label, an estimated fair-value price range and the supporting average are displayed to the Home Buyer.
Priority:	High
Frequency of Use:	Frequent. Used whenever a Home Buyer wants pricing context for a flat.
Flow of Events:	1. Home Buyer requests fair-value pricing for the selected flat. 2. System retrieves the stored rolling 6-month estate average for the flat's town and flat type. 3. System compares the flat's price with the average and assigns a label of Under-Valued, Fair-Valued or Over-Valued (FR-9). 4. System computes an estimated fair-value price range. 5. System displays the label, the estimated range, the average used and the date of the baseline, clearly marked as an estimate.
Alternative Flows:	UC-05.AC.1: Home Buyer may continue to Calculate CPF Grant and Affordability (UC-06) using the displayed price.
Exceptions:	UC-05.EX.1: If there are not enough recent transactions for the flat's town and flat type, the system tells the Home Buyer that a fair-value estimate cannot be computed. UC-05.EX.2: If the last batch refresh failed and the baseline is older than 24 hours, the system still shows the estimate together with the baseline date and a notice.
Includes:	None. Extends: View Flat Details (UC-04).
Special Requirements:	Pricing estimates must be clearly labelled as estimates, not guarantees. The live data.gov.sg API is not called here. The stored baseline is used so the page stays fast.
Assumptions:	The valuation baseline is kept current by Update Valuation Baseline (UC-12).
Notes and Issues:	None at this time.

Use Case ID:	UC-06		
Use Case Name:	Calculate CPF Grant and Affordability		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	6 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Estimates the CPF housing grants (EHG and PHG) the Home Buyer may receive for a selected flat and the resulting monthly cash outlay. This use case extends View Flat Details.
Preconditions:	Home Buyer is viewing the details of a selected flat.
Postconditions:	Estimated EHG and PHG amounts, the monthly cash outlay, an affordability rating and the loan assumptions used are displayed to the Home Buyer.
Priority:	High
Frequency of Use:	Frequent. Used by most Home Buyers evaluating a flat.
Flow of Events:	1. Home Buyer opens the CPF Grant and Affordability section. 2. System invokes Input Financial Profile (UC-07) to obtain the Home Buyer's financial details. 3. System estimates the EHG from monthly household income and marital status, and the PHG from marital status and how close the parents live to the flat (FR-11). 4. System calculates the monthly cash outlay using the flat price, the grants and the default loan assumptions. 5. System displays the grants, monthly cash outlay, affordability rating and loan assumptions (FR-12), together with a disclaimer that final amounts are decided by HDB.
Alternative Flows:	UC-06.AC.1: Home Buyer changes the loan tenure or down payment and the system recalculates the monthly cash outlay (FR-13). UC-06.AC.2: Home Buyer updates the financial profile and the flow resumes at step 2.
Exceptions:	UC-06.EX.1: If the financial profile is incomplete, the system asks the Home Buyer for the missing information before calculating. UC-06.EX.2: If the Home Buyer is not eligible for a grant, the system shows $0 for that grant together with the reason.
Includes:	Includes: Input Financial Profile (UC-07). Extends: View Flat Details (UC-04).
Special Requirements:	Grant rules are stored in a configuration file so they can be updated without changing code. Financial data must be protected as described in NFR-6.
Assumptions:	The grant rules in the configuration file match the latest HDB rules.
Notes and Issues:	CPF grant rules should be reviewed whenever HDB announces changes. In version 1.1 this use case extended View Fair-Value

	Pricing. It now extends View Flat Details, because the CPF calculator sits directly on the Flat Detail screen.

Use Case ID:	UC-07		
Use Case Name:	Input Financial Profile		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	6 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Collects the financial details needed to estimate CPF grants and affordability: monthly household income, marital status and parental address. This use case is included by Calculate CPF Grant and Affordability.
Preconditions:	Home Buyer has started a CPF Grant and Affordability calculation, or chose to edit the financial profile from the Profile screen.
Postconditions:	A valid financial profile is available to the calling use case. If the Home Buyer is signed in and chose to save it, the profile is stored in encrypted form.
Priority:	High
Frequency of Use:	Once per session, unless updated.
Flow of Events:	1. System presents the financial profile form. 2. Home Buyer enters monthly household income, marital status and parental address (with postal code). 3. System validates the fields. 4. System works out the distance between the parental address and the selected flat for the PHG check. 5. If the Home Buyer is signed in and ticks “Save for next time”, system stores the profile in encrypted form. 6. System returns control to the calling use case.
Alternative Flows:	UC-07.AC.1: A signed-in Home Buyer with a saved profile sees the form already filled in and can confirm it without re-entering details. UC-07.AC.2: A guest Home Buyer’s profile is kept only for the current session and is never stored.
Exceptions:	UC-07.EX.1: If required fields are missing or invalid, the system shows a validation error and asks for correction. UC-07.EX.2: If the parental postal code cannot be found, the system asks the Home Buyer to correct it or to skip the PHG estimate.
Includes:	None.
Special Requirements:	Income and parental address are encrypted at rest with AES-256 and sent over TLS 1.3. They must never appear in server logs or be sent to external APIs (NFR-6). For this reason the parental postal code is looked up in the system’s own postal code table rather than through OneMap.
Assumptions:	Home Buyer provides reasonably accurate financial information.
Notes and Issues:	None at this time.

Use Case ID:	UC-08		
Use Case Name:	Bookmark Flat		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	6 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Allows the Home Buyer to save a flat of interest to their personal bookmark list for later reference, or remove it again.
Preconditions:	Home Buyer is viewing a flat in the search results or on the Flat Detail screen.
Postconditions:	The selected flat is added to (or removed from) the Home Buyer's bookmark list, and the change persists across sessions.
Priority:	Medium
Frequency of Use:	Frequent. Used for flats of interest during a session.
Flow of Events:	1. Home Buyer selects the bookmark option for a flat. 2. If there is no active signed-in session, system invokes Authenticate User (UC-11). 3. System checks that the flat is not already bookmarked. 4. System saves the bookmark against the Home Buyer's account. 5. System confirms that the flat has been bookmarked (the star icon is filled in).
Alternative Flows:	UC-08.AC.1: If the flat is already bookmarked, selecting the option again removes it from the bookmark list (FR-15).
Exceptions:	UC-08.EX.1: If a duplicate bookmark is attempted (for example, from two open tabs), the system tells the Home Buyer and does not create a second entry. UC-08.EX.2: If the bookmark cannot be saved, the system shows an error and leaves the bookmark list unchanged.
Includes:	Includes: Authenticate User (UC-11).
Special Requirements:	Bookmarks must persist across sessions (FR-21).
Assumptions:	The Home Buyer is identified across sessions through their account.
Notes and Issues:	Version 1.1 stated that this use case extends Compare Bookmarked Flats. That relationship was removed (see UC-09).

Use Case ID:	UC-09		
Use Case Name:	Compare Bookmarked Flats		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	6 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Allows the Home Buyer to compare two or more bookmarked flats side by side on price, commute, livability and valuation.
Preconditions:	Home Buyer has bookmarked two or more flats.
Postconditions:	A side-by-side comparison of the selected flats is displayed, with differences highlighted.
Priority:	Medium
Frequency of Use:	Occasional. Used when narrowing down a shortlist.
Flow of Events:	1. Home Buyer opens the Bookmarks screen. If not signed in, system invokes Authenticate User (UC-11). 2. Home Buyer selects two or more bookmarked flats. 3. Home Buyer selects “Compare”. 4. System retrieves the current price, commute times, Livability Index and valuation label for each selected flat. 5. System displays the flats side by side and highlights where they differ.
Alternative Flows:	UC-09.AC.1: Home Buyer removes a flat from the comparison. The comparison stays open as long as at least two flats remain. UC-09.AC.2: Home Buyer starts from the “Compare” button on the Flat Detail screen. The system opens the Bookmarks screen with that flat already selected.
Exceptions:	UC-09.EX.1: If fewer than two flats are selected, the system asks the Home Buyer to select at least two.
Includes:	Includes: Authenticate User (UC-11).
Special Requirements:	Comparison should clearly highlight differences between flats (price, commute time, Livability Index and valuation).
Assumptions:	Commute times use the anchors from the Home Buyer’s most recent search. If there has been no search this session, commute is shown as not available.
Notes and Issues:	In version 1.1 this use case was drawn as an extension of Bookmark Flat. Comparing is a separate goal that the Home Buyer starts on purpose, so it is now a standalone use case that needs bookmarks as a precondition.

Use Case ID:	UC-10		
Use Case Name:	Filter Search Results		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	21 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Allows the Home Buyer to narrow down the ranked list from Search Flats by Dual-Anchor Commute using extra criteria: price range, flat type and town. This use case extends Search Flats by Dual-Anchor Commute.
Preconditions:	Home Buyer has performed a flat search and a ranked result list is displayed.
Postconditions:	The displayed list shows only flats that match the filter criteria, in the same ranked order.
Priority:	Medium
Frequency of Use:	Frequent. Used whenever the initial result set is too broad.
Flow of Events:	1. Home Buyer selects the filter option on the Results screen. 2. System displays the available filter criteria (price range, flat type and town). 3. Home Buyer selects one or more filter values. 4. System applies the filters to the current ranked result set. 5. System displays the filtered, still-ranked list of flats.
Alternative Flows:	UC-10.AC.1: Home Buyer clears all filters to return to the full ranked list.
Exceptions:	UC-10.EX.1: If no flats match the filters, the system tells the Home Buyer and suggests which filter to relax.
Includes:	None. Extends: Search Flats by Dual-Anchor Commute (UC-01).
Special Requirements:	Filtering is done on the stored result set without calling the Data APIs again, so it stays within the NFR-1 target.
Assumptions:	Price, flat type and town are available for every flat in the result set.
Notes and Issues:	Changed from an include to an extend relationship, because filtering is optional and only happens when the Home Buyer asks for it. Floor level and amenities were removed as filters to match FR-18 and FR-22 in the SRS.

Use Case ID:	UC-11		
Use Case Name:	Authenticate User		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	21 September 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Verifies the Home Buyer's identity before allowing access to features that need a personal, saved record: bookmarks, comparison and profile. This use case is included by Bookmark Flat, Compare Bookmarked Flats and Manage Profile.
Preconditions:	Home Buyer has tried to use a feature that needs a signed-in session, and no valid session exists.
Postconditions:	The Home Buyer's identity is verified and a time-limited session is created, or access is denied.
Priority:	High
Frequency of Use:	Once per session, or whenever the session has expired.
Flow of Events:	1. System detects that the requested action needs a signed-in session. 2. System displays the Login screen. 3. Home Buyer enters email and password and submits. 4. System checks the password against the stored salted hash. 5. System creates a session and issues a signed, time-limited access token (JWT). 6. System returns control to the calling use case, which then completes the original action.
Alternative Flows:	UC-11.AC.1: Home Buyer has no account and selects "Sign Up". The flow continues in Register Account (UC-13). UC-11.AC.2: Home Buyer cancels and returns to the previous screen without completing the action.
Exceptions:	UC-11.EX.1: If the credentials are invalid, the system shows a general error (without saying which field was wrong) and lets the Home Buyer try again. UC-11.EX.2: After a configurable number of consecutive failed attempts, the system locks the account for a set time and informs the Home Buyer (FR-26).
Includes:	None. Extended by: Register Account (UC-13).
Special Requirements:	Passwords are stored only as salted hashes. Session tokens are sent over TLS 1.3 and expire after a period of inactivity.
Assumptions:	The Home Buyer has a registered account, or registers one through UC-13.
Notes and Issues:	The number of failed attempts before lockout and the session timeout are listed as TBD-4 and TBD-5 in the SRS.

Use Case ID:	UC-12		
Use Case Name:	Update Valuation Baseline (24h Batch Refresh)		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	21 September 2026	Date Last Updated:	1 October 2026

Actor:	Scheduler (Batch Job), Data APIs (data.gov.sg)
Description:	Automatically refreshes the valuation baseline used by View Fair-Value Pricing by pulling the latest resale transactions from data.gov.sg every 24 hours.
Preconditions:	The batch scheduler is running and 24 hours have passed since the last successful refresh.
Postconditions:	The rolling 6-month estate averages are updated with the latest transactions. A record of the batch run (time, status and number of records) is stored.
Priority:	Medium
Frequency of Use:	Automatic. Once every 24 hours.
Flow of Events:	1. Scheduler triggers the batch refresh job at the scheduled time. 2. System queries data.gov.sg for new or updated resale transactions since the last successful refresh. 3. System stores the new transactions. 4. System recomputes the rolling 6-month estate average for each affected town and flat type. 5. System stores the updated baseline and records the refresh time and number of records ingested.
Alternative Flows:	UC-12.AC.1: When the server starts, the start-up class checks the date of the last successful refresh. If it is more than 24 hours old, the refresh runs immediately.
Exceptions:	UC-12.EX.1: If data.gov.sg is unavailable or returns an error, the system keeps the previous baseline, logs the failed run and raises an internal alert (FR-30). UC-12.EX.2: If the retrieved data is incomplete, the system flags the affected towns and does not apply a partial update to them.
Includes:	None.
Special Requirements:	The refresh must not slow down Home Buyer requests (NFR-1). It runs in the background at a low-traffic time.
Assumptions:	data.gov.sg publishes updated transaction data regularly.
Notes and Issues:	None at this time.

Use Case ID:	UC-13		
Use Case Name:	Register Account		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	1 October 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Allows a Home Buyer without an account to create one so they can use bookmarks, comparison and profile features. This use case extends Authenticate User.
Preconditions:	Home Buyer is on the Login screen and has no existing account.
Postconditions:	A new account is created, the Home Buyer is signed in, and the original action continues.
Priority:	High
Frequency of Use:	Once per Home Buyer.
Flow of Events:	1. Home Buyer selects “Sign Up” on the Login screen. 2. System displays the sign-up form with email, password, confirm password and a personal data consent notice. 3. Home Buyer enters the details and accepts the consent notice. 4. System checks the email format, that the email is not already registered, and that the password is strong enough. 5. System creates the account and stores the password as a salted hash. 6. System signs the Home Buyer in and returns to the action that triggered authentication (FR-27).
Alternative Flows:	UC-13.AC.1: Home Buyer goes back to the Login screen without registering.
Exceptions:	UC-13.EX.1: If the email is already registered, the system says so and offers to go to the Login screen. UC-13.EX.2: If the password is too weak or the two passwords do not match, the system shows an error. UC-13.EX.3: If the consent notice is not accepted, the account is not created.
Includes:	None. Extends: Authenticate User (UC-11).
Special Requirements:	Consent must be recorded to meet PDPA requirements.
Assumptions:	Home Buyer has a valid email address.
Notes and Issues:	This was an alternative flow (UC-11.AC.1) in version 1.1. It is now its own use case because it has its own steps, rules and exceptions.

Use Case ID:	UC-14		
Use Case Name:	Manage Profile		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	1 October 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Allows a signed-in Home Buyer to view their account details and saved preferences, and to update or delete their saved financial profile.
Preconditions:	Home Buyer selects the Profile tab on the navigation bar.
Postconditions:	The Home Buyer's saved profile data reflects any changes they made.
Priority:	Medium
Frequency of Use:	Occasional.
Flow of Events:	1. Home Buyer selects the Profile tab. If not signed in, system invokes Authenticate User (UC-11). 2. System displays the account email, the saved weight profile and a summary of the saved financial profile. 3. Home Buyer chooses to edit or delete the saved financial profile. 4. System validates and saves the changes, or deletes the stored financial data. 5. System confirms the change.
Alternative Flows:	UC-14.AC.1: Home Buyer resets the saved weight profile to the system defaults.
Exceptions:	UC-14.EX.1: If the edited financial details are invalid, the system shows a validation error and keeps the previous values.
Includes:	Includes: Authenticate User (UC-11).
Special Requirements:	Deleting the financial profile must remove the data from the database, to support PDPA requirements.
Assumptions:	None.
Notes and Issues:	None at this time.

Use Case ID:	UC-15		
Use Case Name:	Sign Out		
Created By:	Team ABCDE	Last Updated By:	Team ABCDE
Date Created:	1 October 2026	Date Last Updated:	1 October 2026

Actor:	Home Buyer
Description:	Ends the Home Buyer's signed-in session.
Preconditions:	Home Buyer is signed in and on the Profile screen.
Postconditions:	The session is ended and the Home Buyer is back on the Search screen as a guest.
Priority:	Medium
Frequency of Use:	Once per session, at most.
Flow of Events:	1. Home Buyer selects "Sign Out" on the Profile screen. 2. System ends the session on the server. 3. System discards the access token and any personal data held in the browser. 4. System returns the Home Buyer to the Search screen as a guest.
Alternative Flows:	UC-15.AC.1: If the session expires because of inactivity, the Home Buyer is treated as signed out. The next protected action triggers Authenticate User (UC-11).
Exceptions:	UC-15.EX.1: If the server cannot be reached, the browser still clears the session locally and the server-side token expires on its own.
Includes:	None.
Special Requirements:	None identified.
Assumptions:	None.
Notes and Issues:	None at this time.